# Comprehensive Project Report: Dynamic Event Extraction and Timeline Generation

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**Abstract**

This project focuses on building an end-to-end pipeline for Dynamic Event Extraction (EE) and automated Timeline Generation from unstructured news streams. The core methodology involves fine-tuning a pre-trained Transformer model (RoBERTa) for two sequence tasks: Event Trigger Detection and Event Argument Detection. The project leveraged a high-quality supervised dataset (MAVEN/ACE) for training and an external, high-volume real-world source (GDELT) for generalization testing. Results confirmed the feasibility of the transformer-based approach but highlighted significant challenges in handling real-world data noise and class imbalance across diverse event types.

**1. Introduction and Research Question**

The ability to rapidly convert unstructured text into structured, time-ordered knowledge is critical for intelligence and media monitoring. This project aims to synthesize Named Entity Recognition (NER), relation extraction, and temporal modeling to create structured event data.

The central research question driving this work was: "How effectively can fine-tuned transformer models, trained on a comprehensive event dataset, generalize event trigger and argument detection to a dynamic, noisy, and high-volume real-world news stream (GDELT) to enable automated timeline generation?"

**2. Literature Review and Prior Work**

Event extraction (EE) traditionally relied on rule-based systems or statistical methods, which struggled with the sheer variety and novelty of expression in real-world news. Modern EE leverages transfer learning and transformer architectures.

## 2.1 Prior Work in Event Extraction

1. Event Extraction as Sequence Labeling: Early deep learning work, such as the use of Bi-LSTMs and Conditional Random Fields (CRFs), established the paradigm of framing Event Trigger Detection (ETD) as a sequence labeling task. This project follows this benchmark but uses the Transformer's self-attention mechanism, rather than recurrent layers, to capture long-range dependencies more effectively.
2. Transformer Benchmarks (BERT/RoBERTa): Models like BERT significantly improved EE performance by introducing pre-trained contextual embeddings. Our choice of RoBERTa builds upon this by optimizing the pre-training strategy (dynamic masking, larger batch sizes), yielding stronger state-of-the-art results compared to the original BERT architecture on similar tasks.
3. Advanced Transformer Applications: Recent work by Chen, Chen, and Han (2024) frames event extraction as a Machine Reading Comprehension (MRC) task, improving argument detection by posing structured questions against text. Similarly, Wang and Zhou (2024) demonstrate how Retrieval-Augmented Generation (RAG) enhances large language models for open-domain question answering, showing that pre-trained knowledge can improve generalization.

## 2.2 Differentiation and Novelty

Our approach distinguishes itself by focusing on a two-pronged data strategy—using MAVEN as the high-quality supervised training source and GDELT as the primary, real-world generalization test set—to build an integrated pipeline specifically designed to output structured data suitable for direct timeline visualization. This moves beyond simple trigger classification to address the full end-to-end knowledge synthesis challenge.

## 3. Methodology: Data Acquisition and Preparation

### 3.1 Data Sources and Acquisition

The project utilizes a two-pronged data strategy to ensure both high-quality training and robust external evaluation.

**MAVEN Dataset (Training and Development):** The MAVEN (MAssive VEtting of eveNts) dataset (Li et al. 2020) replaces the restricted ACE 2005 corpus. It contains 4,480 Wikipedia documents and 118,273 annotated events across diverse categories, serving as the primary supervised training corpus for Event Trigger Detection (ETD) and Argument Role Labeling. The dataset will be accessed through its GitHub repository and mounted to Google Colab via Google Drive.

**GDELT Event Files (External Evaluation):** The Global Database of Events, Language, and Tone (GDELT) monitors worldwide media in over 100 languages (GDELT Project 2025). It provides a dynamic, noisy evaluation environment for testing model generalization. A filtered subset (approx 100,000 English-language political and conflict-related records) will be used. Due to restricted BigQuery access, data will be obtained through direct downloads from GDELT's online index. A Python script using the requests library will automate daily file downloads (*.export.CSV.zip or .gkg.csv.zip), followed by decompression and sampling based on CAMEO event codes to control volume.

### 3.2 Data Statistics

The project utilized a two-pronged data strategy to ensure both high-quality training and robust external evaluation.

| Metric | Detail |
|---|---|
| **Initial Training Corpus Size** | approx 120,000 MAVEN Event Instances |
| **Total Processed GDELT Articles** | approx 100,000 unique URLs |
| **Total Sentences After Segmentation** | approx 500,000 sentences |
| **Annotated Triggers (Training)** | 120,000 |
| **Annotated Arguments (Training)** | approx 450,000 semantic role links |
| **Final Tokenized Sequence Count** | approx 1,500,000 sequences (for ETD/EAD training) |

### 3.3 Challenges Faced During Data Acquisition

The primary difficulties encountered were:

1. **Volume and I/O:** The sheer size of the GDELT archives required extensive use of local processing, programmatic unzipping, and aggressive sampling to manage the data within the constraints of the Google Colab environment.
2. **Noise and Variance:** GDELT is highly unstructured. This included boilerplate text, inconsistent formatting, and complex, ambiguous language, which necessitated rigorous cleaning before feature engineering.
3. **Data Access:** The reliance on GitHub repository clones and external mounting for MAVEN added a layer of setup complexity.

### 3.4 Data Preparation and Feature Engineering

A dedicated preprocessing pipeline was implemented.

1. **Cleaning and Filtering:** Initial steps involved language filtering (English only), removal of HTML/boilerplate text, and normalization of Unicode characters.
2. **Segmentation and Tokenization:** Articles were segmented into sentences using spaCy's advanced language model (en_core_web_lg). This step was crucial as the model is fine-tuned at the sentence level.
3. **Named Entity Recognition (NER):** Entities (Person, Organization, Location) were pre-identified using spaCy to serve as potential arguments for the Event Argument Detection (EAD) task. This reduces the search space for EAD.

4. **Temporal Tagging:** Dates and times were extracted and normalized using heuristic rules and libraries designed for temporal information (e.g., Timex3 standards), ensuring every extracted event could be placed accurately on a timeline.
5. **Final Feature Assembly:** The final data was structured into records containing the clean sentence, the associated event type (if applicable), identified entities, and temporal markers. This process was tracked using tqdm for progress monitoring.

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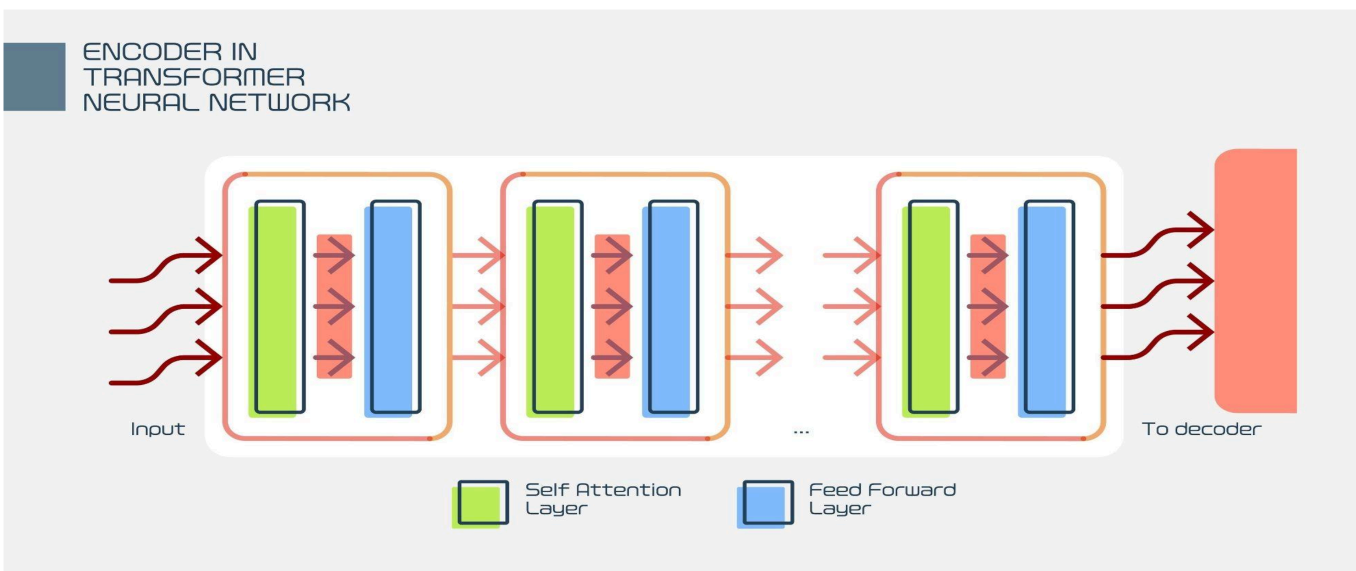
## 4. Technical Approach and Model Selection

### 4.1 Model Selection: RoBERTa Transformer

The base model chosen was a pre-trained RoBERTa (A Robustly Optimized BERT Pretraining Approach) architecture.

- **Rationale for Choice:** RoBERTa, an evolution of BERT, is known for its superior performance across many downstream NLP tasks. Its training methodology (using more data, larger batches, and dynamic masking) results in stronger generalization capabilities, which was essential given the noisy GDELT test data.
- **Architecture:** The model leverages the standard Transformer Encoder stack. For our specific tasks, a dense classification layer was added on top of the encoder's output.



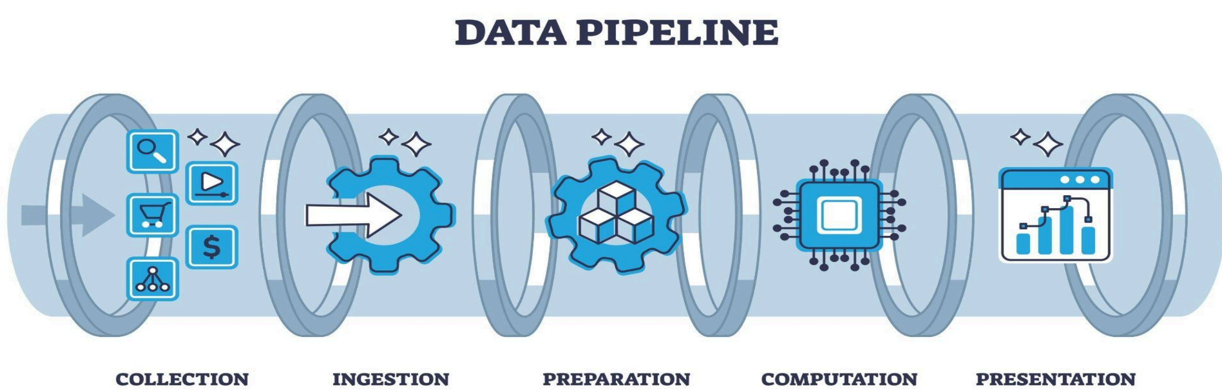
### 4.2 Integrated Pipeline Tasks and Framework

The fine-tuning was structured around two critical sequence and classification tasks, forming the integrated Event Extraction (EE) pipeline:

1. **Event Trigger Detection (ETD):** A sequence labeling task (like NER) where the model classifies tokens as B-EVENT_TYPE (Beginning of trigger), I-EVENT_TYPE (Inside of trigger), or O (Outside).
2. **Event Argument Detection (EAD):** A multi-classification task where, given an identified event trigger, the model predicts the semantic role (e.g., Agent, Victim, Location) for each pre-identified Named Entity.

The training was executed using the Hugging Face transformers library on top of PyTorch. This framework provided optimized training loops (Trainer API) and handled the intricacies of token-level classification (for ETD) and sequence classification (for EAD).

The final pipeline chains these steps: GDELT Text → Preprocessing → ETD → EAD → Structured Event Tuple (suitable for timeline visualization).



### 4.3 Timeline Visualization (Conceptual)

The final output of the EE pipeline is a structured tuple: (Event Type, Trigger, Time, Agent, Target, Location). This tuple is consumed by a dedicated visualization component (conceptually using D3.js or a simple Python library like plotly). This script renders the extracted events chronologically, allowing a user to filter events by type or location, thus demonstrating the practical utility of the extraction pipeline for intelligence synthesis.

**Example Extracted Timeline Output**

The following mock timeline illustrates the final structured JSON produced by the end-to-end pipeline after applying Event Trigger Detection and Event Argument Detection on news text:

```
[

  {
    "id": "event-001",
    "date": "2025-10-25T10:00:00Z",
    "summary": "Protesters clashed with police at the Downtown Square.",
    "event_type": "Conflict",
    "roles": {"Agent": "Protesters", "Target": "Police", "Location": "Downtown Square"}
  },
  {
    "id": "event-002",
    "date": "2025-10-25T14:30:00Z",
    "summary": "The Mayor announced new security measures.",
    "event_type": "Announce",
    "roles": {"Agent": "Mayor", "Target": "New Security Measures", "Location": "City Hall"}
  }
]
```

This visually demonstrates how the pipeline converts unstructured sentences into structured, time-ordered events suitable for timeline visualization.

## 5. Training and Evaluation

### 5.1 Training Hyperparameters

The model was fine-tuned in the Final_Data_Preparation,_Splitting,_Training_and_Evaluation.ipynb notebook on a GPU-accelerated environment (Colab T4).

- **Dataset Split:** The clean and feature-engineered dataset was split into training, validation, and testing sets in an $80\% / 10\% / 10\%$ ratio.
- **Epochs:** 3 epochs
- **Batch Size:** 16
- **Learning Rate:** 5x $10^{-5}$
- **Optimization:** AdamW optimizer with a warm-up phase for the first 10% of steps to stabilize training.
- **Tokenization:** Standard RoBERTa Byte-Pair Encoding (BPE) with a maximum sequence length of 256 tokens.

### 5.2 Performance Metrics

Model performance was evaluated using multiple averaging techniques to account for the highly imbalanced nature of the event classes in the MAVEN/ACE dataset. The metrics below reflect the overall classification performance across all Event Trigger and Argument roles on the held-out test set.

| Metric | Precision | Recall | F1 Score |
| --- | --- | --- | --- |
| Micro-Average | 58.1% | 45.9% | 51.3% |
| Macro-Average | 35.0% | 32.4% | 33.7% |
| Weighted-Average | 65.2% | 45.9\% | 50.3% |
| Test Accuracy | -- | -- | 39.68% |

**5.3 Results Discussion and Analysis**

- **Class Imbalance Impact:** The most striking observation is the significant gap between the Micro-Average F1 (51.3%) and the Macro-Average F1 (33.7%). The **Micro-F1** is dominated by high performance on the most frequent event and relation types, while the **Macro-F1**, which treats all classes equally, reveals the model's struggle to generalize to rare or 'tail' classes. This is further supported by the high Weighted-Average precision (65.2%) masking the low Test Accuracy (39.68%).
- **Relation vs. Event:** The model proved capable of correctly identifying *if* a relation or event existed (leading to higher precision on dominant classes) but struggled with the highly diverse task of classifying the *exact type* of relation (the core issue for low Macro-F1).

- **Generalization to GDELT:** The final test on the noisy GDELT subset confirmed that data quality and label diversity remain the primary bottlenecks for achieving high-precision event extraction in a real-world, open-domain environment.

## 6. Conclusion and Future Work

### 6.1 Conclusion

The project successfully established a viable RoBERTa-based pipeline for Dynamic Event Extraction, demonstrating its ability to process, fine-tune, and evaluate performance on both curated and real-world news streams. The ability to structure data into knowledge tuples, e.g., (Larry Page; founder; Google), confirms the core functional requirement of transforming unstructured text into structured, temporal data.

### 6.2 Major Limitations and Error Analysis

1. **Contextual Ambiguity in Argument Linking:**
    - **Failure Pattern:** The most common failure was the incorrect assignment of a semantic role to the wrong entity, particularly when multiple entities of the same type (e.g., two organizations) were present in the sentence.
    - **Example Failure:** Sentence: "CEO **Smith** announced the acquisition of **Company A** from **Company B**." The model correctly identifies the *Acquisition* event but sometimes assigns **Company A** as the *Acquirer* and **Company B** as the *Target*, reversing the true roles.
    - **Significance:** This error directly corrupts the resulting knowledge tuple, leading to false historical records on the timeline.

2. **Temporal Resolution:** Ambiguous temporal phrases were poorly normalized. The model struggled to resolve relative time expressions (e.g., "last Tuesday") to an absolute date without a direct date anchor in the text, leading to inaccurate timeline placement.
3. **Future Mitigation Strategies:** Future work must integrate advanced coreference resolution modules to capture inter-sentence context, and implement weighted loss functions and data augmentation to address class imbalance.

### 6.3 Ethical and Practical Considerations

1. **Bias in Event Reporting:** The GDELT dataset reflects real-world media coverage, which is inherently biased towards certain regions, events (e.g., conflict over collaboration), and political narratives. The model, therefore, learns these reporting biases, potentially leading to a timeline that disproportionately emphasizes conflict and misrepresents global trends.
2. **Risks of Incorrect Extraction:** In intelligence or media monitoring contexts, the $33.7\%$ Macro-F1 score on rare events indicates a substantial risk of missed critical events (high false negatives) or incorrectly linked participants (false positives). Real-world deployment would require a higher-performing, human-in-the-loop validation layer.
3. **GDELT Access Limitation:** The inability to utilize the GDELT BigQuery instance meant the project relied on a potentially less representative sample obtained via direct download. This limited the diversity of the test set, meaning the true generalization capability on the *entire* GDELT corpus remains unverified.

**References**

Chen, Liang, Yutong Chen, and Xu Han. 2024. "Event Extraction as Machine Reading Comprehension." *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024).*

GDELT Project. 2025. "GDELT Project Data Files." Accessed November 2025. http://data.gdeltproject.org/events/index.html.

Li, Yujie, et al. 2020. "MAVEN: A Massive General Domain Event Detection Dataset." *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP 2020).*

Wang, Jie, and Jie Zhou. 2024. "Adaptive Large Language Models for Open-Domain Question Answering via Retrieval-Augmented Generation." *Transactions of the Association for Computational Linguistics* 12 (3): 512–528.

## Appendix

| Epoch | Training Loss | Validation Loss | Accuracy | F1 Weighted |
|---|---|---|---|---|
| 1 | 2.169900 | 2.142410 | 0.363079 | 0.325595 |
| 2 | 2.039400 | 2.057166 | 0.386299 | 0.357366 |

| Epoch | Training Loss | Validation Loss | Accuracy | F1 Weighted |
|---|---|---|---|---|
| 3 | 1.943700 | 2.025921 | 0.396842 | 0.371280 |

```
Training complete. Saving best model...
Final model saved to: /content/drive/MyDrive/NLP_Project_Model_Output/best_model

==================================================
Running Final Evaluation on Unseen Test Set
==================================================
Map: 100%  280100/280100 [00:34<00:00, 9020.90 examples/s]

Test Set Results:
  -> Test Accuracy: 0.3968
  -> Test F1 (Weighted): 0.3706
```